

3.5c

I don't know

3.5(d)

When only one village is treated, the estimate depends heavily on that single village's treatment outcome.

The treatment potential outcomes have high variance: $\text{Var}(Y(1))=42.86$

Therefore, selecting a different treatment village can greatly change the estimated ATE. The standard error when one village is treated is: $\text{SE}=6.76$. When two villages are treated: $\text{SE}=4.60$. Thus, treating two villages produces less sampling variability because the treatment mean is based on two observations rather than one.

3.5(e)

The wording of this question appears to be reversed. For the values in Table 2.1, assigning one village to treatment has more, not less, sampling variability than assigning six villages to treatment. With one treated village: $\text{SE}=6.76$ With six treated villages: $\text{SE}=4.23$. This happens because $(Y(1))$ has greater variance than $(Y(0))$. When only one village is treated, the treatment mean relies on one highly variable observation. When six villages are treated, their treatment outcomes are averaged, reducing variability.